Prob. 5. (2 & T. Ch. 7, 7.6)

Consider the Dirichlet problem for the Laplace eq-n where Ω is the unit disk, $0 \le r < 1$, $-\pi \le \theta \le \pi$, & the boundary condition given by $u(1, \theta) = |\theta|$, $-\pi \le \theta \le \pi$. The sol-n is derived in the text (you do NOT have to redirive the sol-n) & is given by

(1) $$u(r,\theta) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi} \cdot \frac{(-1)^n - 1}{n^2} \cdot r^n \cos(n\theta), \quad 0 \le r < 1$$

Use the series sol-n to determine u at par-r loc-ns. Note that this sol-n is <u>only valid</u> for $r < 1$, so we would expect the series to converge more slowly as r approaches 1. Calculate u at the ↓ loc-ns to <u>two decimal</u> places of accuracy, & note the number of non-zero terms that give us this sol-n.

Sol-n.

(a) $u(0, \theta) = ?$

$$u(0,\theta) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi} \cdot \cdots \cdot 0^n \cdot \cos(n\theta) = \frac{\pi}{2}.$$

(2) $\Rightarrow \boxed{u(0,\theta) = \frac{\pi}{2}}$

Let $n = 2k-1$

(b) $$u(0.1, 0) = \frac{\pi}{2} + \sum_{k=1}^{\infty} \frac{2}{\pi} \cdot \frac{(-1)^{2k-1} - 1}{(2k-1)^2} \cdot (0.1)^{(2k-1)} \cdot \cos(n\theta) =$$

$$= \frac{\pi}{2} + \frac{2}{\pi} \cdot \frac{(-1)^1 - 1}{(1)^2} \cdot (0.1)^1 \cdot \cos(\theta) + \frac{2}{\pi} \cdot \frac{(-1)^2 - 1}{(2)^2} (0.1)^2 \cos(2\theta)$$

$$+ \frac{2}{\pi} \cdot \frac{(-1)^3 - 1}{(3)^2} \cdot \underbrace{(0.1)^3}_{\text{3decimal}} \cdot \cos(3\theta) = \frac{\pi}{2} - \frac{0.4}{9\pi} + (\text{3decimal}) =$$

$$\approx \frac{3.14}{2} - 0.127 = 1.57 - 0.13 = 1.44$$ (Needed 2 nonzero terms)

(c) $$u(0.5, 0) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi} \cdot \frac{(-1)^n - 1}{n^2} (0.5)^n \cdot \underbrace{\cos(n\theta)}_{1} =$$

$$= \frac{\pi}{2} + \frac{2}{\pi} \cdot \frac{(-1)^1 - 1}{(1)^2} \cdot (0.5)^1 + \frac{2}{\pi} \cdot \frac{(-1)^2 - 1}{(2)^2} \cdot (0.5)^2 + \frac{2}{\pi} \cdot \frac{(-1)^3 - 1}{(3)^2} (0.5)^3$$

$$= \frac{\pi}{2} + \frac{(-2)}{\pi \cdot 9} \cdot \frac{1}{2^2} = \frac{\pi}{2} - \frac{1}{18\pi} \approx 1.57 - \underset{k=1}{0.6366} - \underset{k=2}{0.017} - 0.0002$$

$= 0.9164.$ (3 nonzero terms needed)

(d) $$u(0.9, 0) = 1.5707 - \overset{k=1}{1.1459} - \overset{k=2}{0.103} - \overset{k=3}{0.03} - \overset{k=4}{0.012} - \underbrace{0.006}_{\text{from here 2 decimal points does not affected}}$$

$= 0.279 \approx 0.28$ (5 nonzero terms need.)

Pr. 5.

What is the sol-n along the diameter $\theta = \pm \frac{\pi}{2}$, then

$$u(r,\theta) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi} \cdot \frac{(-1)^n - 1}{n^2} r^n \cos\left(\pm \frac{\pi n}{2}\right)$$

- $\pm n = \pm 4k$, $\cos\left(\pm \frac{4k\pi}{2}\right) = \cos(\pm 2\pi k) = 1$, $\Sigma = 0 \Rightarrow u = \frac{\pi}{2}$
- $\pm n = \pm(4k+1)$, $\cos\left(\pm \frac{(4k+1)\pi}{2}\right) = 0$, $\Sigma = 0$, $\Rightarrow u = \frac{\pi}{2}$
- $\pm n = \pm 4k+2$, $(-1)^{4k+2} - 1 = 0$, $\Sigma = 0 \Rightarrow u = \frac{\pi}{2}$
- $\pm n = 4k+3$, $\cos\left(\pm \frac{(4k+3)\pi}{2}\right) = 0$, $\Sigma = 0$, $\Rightarrow u = \frac{\pi}{2}$

$\Rightarrow \boxed{u = \frac{\pi}{2}}$ if $\theta = \pm \frac{\pi}{2}$